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Sweat glands consist of two parts: the deep, convoluted secretory coil found in the dermis and the secretory duct, which reaches from the secretory coil to the skin surface. New technology has allowed the three-dimensional study of these sweat glands, which was previously not possible due to the complex structure and location of the glands.

Aging is known to compromise the function of eccrine sweat glands, which can lead to loss of heat tolerance and increased risk of heatstroke. As aging occurs, people show reduced responses to heat stimulation and to stimulation from the nervous system. The ability to visualize sweat glands in a three-dimensional manner has allowed the study of sweat glands in people of different ages.

To examine changes that occur in sweat gland morphology due to age, researchers studied eccrine sweat glands in a group of young subjects (average age = 33.4 years) and a group of aged subjects (average age = 65.1 years). Average eccrine sweat gland density, sweat gland volume, secretory duct depth, and secretory duct length were determined in both groups (Table 1).

Table 1 Changes in Eccrine Sweat Glands in Response to Age: Mean Scores and Standard Error

Variables	Young group	Aged group
Average sweat gland density (number/mm ²)	0.95 ± 0.1	0.94 ± 0.05
Average sweat gland volume (µm ³ /sweat gland)	1.45 × 10 ⁷ ± 0.15 × 10 ⁷	1.25 × 10 ⁷ ± 0.15 × 10 ⁷
Average secretory duct depth (µm)	1,100 ± 75	750 ± 75
Average secretory duct length (µm)	2,000 ± 100	1,850 ± 100

In another study of human skin, it was found that the average number of pilosebaceous units (PSUs) was approximately 0.30 units per square millimeter. Which of the following statements best describes the relative densities of sweat glands and PSUs in an area of skin?

- ☐ A. Sweat glands are one-third as dense as PSUs. [6%]
- ✓ ☒ B. Sweat glands are three times as dense as PSUs. [90%]
- ☐ C. Sweat glands are one-fourth as dense as PSUs. [1%]
- ☐ D. Sweat glands are four times as dense as PSUs.

Correct

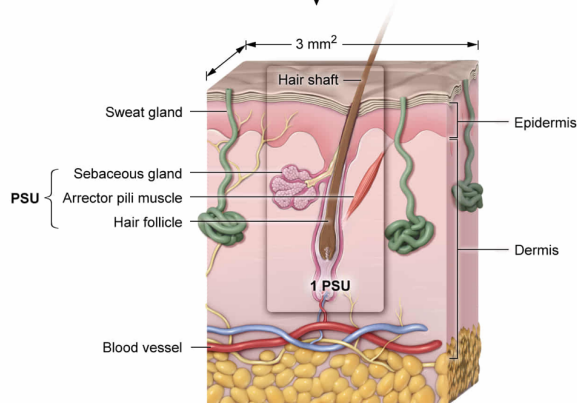
90% answered correctly

Explanation:

Densities of sweat glands and PSUs in human skin

Given PSU density (from question) ≈ 0.30 PSUs/mm²
Given sweat gland density (from Table 1) ≈ 0.945 glands/mm²

↓ Visualization of 3 mm² section of skin



Section contains 1 PSU and 3 sweat glands.
Therefore, sweat glands are three times as dense as PSUs.

PSU = pilosebaceous unit.

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Pilosebaceous units (PSUs) are defined as hair follicles and the sebaceous glands associated with these follicles. While both **eccrine sweat glands** and sebaceous glands are **exocrine glands**, eccrine sweat glands produce fluid **sweat** (a hypotonic fluid), and the sebaceous glands of the PSU produce **sebum** (an oil), which helps prevent water loss from the **skin** and hair.

This question states that in another study of human skin, it was found that the average number of PSUs was approximately 0.30 units/mm². The question asks for the statement that best describes the relative densities of sweat glands and PSUs in an area of skin. The passage table provides the average sweat gland density in both the young and aged groups. Averaging these densities gives an overall average eccrine sweat gland density of 0.945

glands/mm².

A comparison of densities of sweat glands and PSUs is calculated as follows:

$$\text{Relative densities of sweat glands and PSUs} = \frac{\text{Sweat gland density}}{\text{PSU density}}$$

$$\text{Relative densities of sweat glands and PSUs} = \frac{0.945 \text{ glands/mm}^2}{0.3 \text{ PSUs/mm}^2} \approx \frac{3 \text{ sweat glands}}{1 \text{ PSU}}$$

Therefore, **sweat glands are approximately three times as dense as PSUs.**

(Choices A, C, and D) These options represent *incorrect* calculations of the relative densities of sweat glands and PSUs.

Educational objective:

Sebaceous glands associated with hair follicles make up pilosebaceous units (PSUs). Like sweat glands, PSUs are exocrine glands that deposit their product (ie, sebum) on the skin surface.

Time spent:0 sec

Subject:
Biology

QID:403949

System:
Skin and Immune Systems

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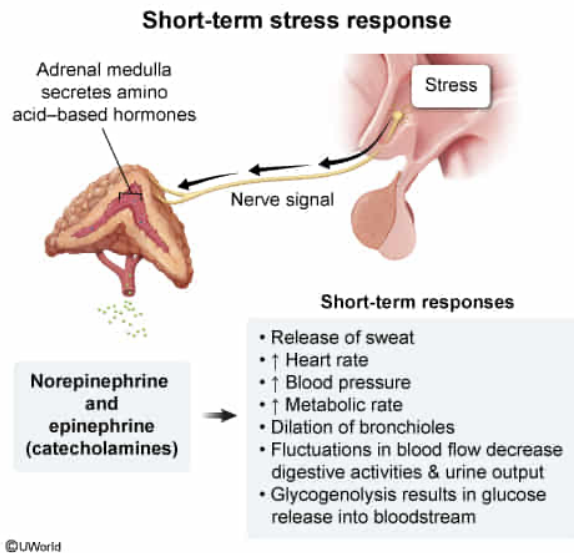
The release of sweat is partially under the control of which hormone?

- ☐ A. Somatostatin [7%]
- ☐ B. Antidiuretic hormone [19%]
- ☐ C. Parathyroid hormone [7%]
- ✓ ☒ D. Epinephrine [65%]

Correct

64% answered correctly

Explanation:



Hormones can affect bodily processes. For example, epinephrine is a hormone **released** by the adrenal medulla in response to a stressor. Epinephrine release leads to redirection of blood flow in the body via vasoconstriction and vasodilation, increased heart rate, dilation of bronchioles, and sweat release. These are hallmarks of the "fight-or-flight" response of the **sympathetic nervous system**.

Therefore, the release of sweat is partially under the control of **epinephrine**.

The other hormones do *not* control the release of sweat.

(Choice A) Somatostatin, released by **pancreatic delta cells**, has a generalized inhibitory effect on the functioning of the digestive system. Somatostatin can also suppress the release of insulin and glucagon from the pancreas.

(Choice B) **Antidiuretic hormone** is released from the **posterior pituitary** to promote water reabsorption by the kidneys when blood pressure falls.

(Choice C) Parathyroid hormone, released from the parathyroid glands in response to low blood calcium levels, works to promote **several responses** to elevate blood calcium levels. If excess calcium losses in sweat caused a reduction in plasma calcium concentration, parathyroid hormone might be released as a *response to* (but not a mediator of) sweating.

Educational objective:

In response to a stressor, the adrenal medulla produces the hormone epinephrine to modulate the "fight-or-flight" response. Some effects of epinephrine include redirection of bodily blood flow, increased heart rate, dilation of bronchioles, and the release of sweat on the skin surface.

Time spent:0 sec

Subject:

Biology

QID:403950

System:

Skin and Immune Systems

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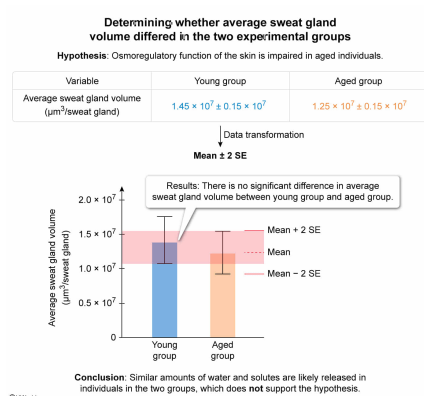
Do the results presented in Table 1 support the hypothesis that the osmoregulatory function of the skin is impaired in aged individuals?

- ☐ A. Yes; the average secretory duct length was significantly shorter in the aged group compared to the young group. [25%]
- ☐ B. Yes; the average sweat gland density was significantly lower in the aged group compared to the young group. [4%]
- ☒ C. No; average sweat gland volume did not differ significantly between the two experimental groups, suggesting that similar amounts of water and solutes are released by individuals in the two groups. [63%]
- ☐ D. No; average secretory duct depth did not differ significantly between the two experimental groups, suggesting that osmoregulation by the skin is not impacted by age. [6%]

Correct

62% answered correctly

Explanation:



Sweat glands in the dermis of the skin secrete sweat, which is a mixture of water, salts, trace nitrogenous waste, and

antimicrobial proteins. Sweat is deposited on the surface of the epidermis via ducts that travel through the dermis and epidermis. Although the **kidneys** are the body's main osmoregulatory organs, skin also functions in an osmoregulatory capacity because sweat contains water and solutes.

This question asks if the hypothesis that the osmoregulatory function of the skin is impaired in aged individuals is supported given the results in Table 1. The table shows that the standard errors for average sweat gland volume **overlapped** (ie, did not differ significantly) between the two experimental groups. Therefore, **aged individuals** likely produce approximately the **same volume** of sweat, on average, **as young individuals**. This suggests that **similar amounts of water and solutes** are released by individuals in the two groups, which **does not support the hypothesis** that the osmoregulatory function of the skin is impaired in aged individuals.

(Choices A and B) The average secretory duct length and average sweat gland density did *not* differ significantly between the two experimental groups.

(Choice D) Given that the average secretory duct depth for the two groups differed by several standard errors, it can be concluded that the secretory duct depth *did* differ significantly between the two experimental groups.

Educational objective:

While the kidneys are the body's main osmoregulatory organs, the skin also plays a role in osmoregulation through the secretion of sweat.

Time spent:0 sec
Subject:
Biology

QID:403951
System:
Skin and Immune Systems

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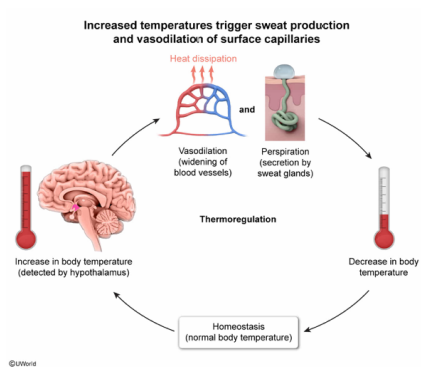
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In addition to sweating, increased temperatures will trigger:

- I. contraction of arrector pili muscles.
 - II. vasodilation of surface capillaries.
 - III. contraction of surface capillary smooth muscle.
- ☐ A. I only [1%]
- ✔ ☒ B. II only [81%]
- ☐ C. I and III only [5%]
- ☐ D. II and III only [11%]

Correct
80% answered correctly

Explanation:



The maintenance of body temperature homeostasis despite changes in ambient temperature and internal heat production is called **thermoregulation**. The hypothalamus contains thermoregulatory centers that monitor body temperature based on information from internal and surface receptors. The **skin** plays a large role in this thermoregulation.

When body temperature increases above normal levels, the hypothalamus coordinates responses to decrease body temperature. These responses include the production of sweat, a hypotonic solution secreted onto the skin surface via glands. When the water in sweat evaporates, heat is lost and the body cools. In addition, surface capillaries undergo **vasodilation**, allowing blood in capillaries to transfer heat from the body to the environment.

When body temperature decreases below normal levels, the hypothalamus coordinates responses to increase body temperature. One response is shivering, which generates heat in the body via contractions of skeletal muscle. In addition, surface capillaries undergo vasoconstriction to minimize heat loss to the environment from the blood.

This question asks what increased temperatures will trigger in the body, in addition to sweating. The other main response to increased temperatures is **vasodilation of surface capillaries** to release heat from the body to the environment **(Number II)**.

(Number I) Contraction of arrector pili muscles causes piloerection, or upright standing of hairs, in response to sympathetic stimulation in cold environments. Therefore, this would *not* be a thermoregulatory response to increased temperatures.

(Number III) Contraction of surface capillary smooth muscle would lead to vasoconstriction, which occurs in decreased (*not* increased) temperatures.

Educational objective:

The skin plays a large role in thermoregulation of the body. Responses that decrease body temperature include sweating and vasodilation of surface capillaries, while responses that increase body temperature include shivering and vasoconstriction of surface capillaries.

Time spent:0 sec
Subject:
Biology

QID:403952
System:
Skin and Immune Systems

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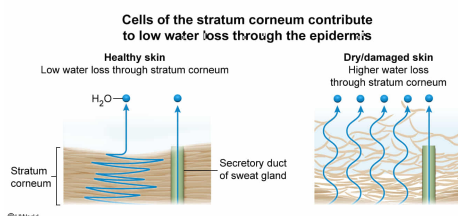
Although some water is lost through sweat glands at the skin surface, the skin is relatively impermeable to water loss largely due to the cells of the:

- ✓ ☒ A. stratum corneum. [42%]
☐ B. stratum basale. [14%]
☐ C. dermis. [26%]
☐ D. hypodermis. [17%]

Correct

42% answered correctly

Explanation:



The **skin** can be divided into three major layers, the **epidermis**, the **dermis**, and the **hypodermis**. The epidermis is further divided into five additional layers known as strata. From superficial to deep, these

layers include the stratum corneum, stratum lucidum (only present in thick skin of the palms and foot soles), stratum granulosum, stratum spinosum, and stratum basale.

This question asks which cells are largely responsible for the skin's relative impermeability to water loss. The most superficial layer of the epidermis, the stratum corneum, is composed of many layers of keratinocytes that have been pushed up from the deeper layers and flattened in the process. These flattened keratinocytes create a barrier to invasion by external pathogens and help prevent water and salt loss through the skin surface. Therefore, the skin is relatively impermeable to water loss largely due to the cells of the **stratum corneum**.

(Choice B) The stratum basale is the deepest layer of the epidermis and consists of a single layer of stem cells that divide continually to give rise to new stem cells. These cells are *not* largely responsible for the skin's relative impermeability.

(Choice C) The dermis is composed of connective tissue and contains sensory receptors, blood vessels, hair follicles, and sweat and sebaceous glands. However, the dermis is *not* primarily responsible for the relative impermeability of the skin.

(Choice D) The hypodermis is composed of adipose tissue and functions in insulation and shock absorption, *not* impermeability to water loss.

Educational objective:

The epidermis is the outermost layer of the skin and acts as a physical barrier to separate the external environment from the body. This barrier contributes to the relative impermeability of the skin to water loss.

Time spent:0 sec
Subject:
Biology

QID:403953
System:
Skin and Immune Systems